BYTECUBIT TECHNOLOGIES · INDUSTRIAL AUTOMATION ENGINEER

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Industrial Systems Modernization | Manufacturing Data & Automation

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Role Alignment Brief

Where my verified experience meets Bytecubit's industrial automation mandate.

THE OPERATING MANDATE

Modernize legacy manufacturing applications with Ignition, strengthen SQL-backed workflows, improve historian data quality, and create a dependable plant-to-cloud path while production continues to run. The work sits at the intersection of software, manufacturing consequence, data context, and cross-functional execution.

ALIGNMENT AT A GLANCE

Manufacturing systems

Vape-Jet and Compunetics provide direct experience across production, engineering, quality, machine systems, issue flow, root cause, and support in technical manufacturing environments.

Modernization discipline

ERP/Odoo implementation, workflow redesign, telemetry, dashboards, standard work, and operating-system development demonstrate how I move fragmented work toward a supportable target state.

Data and integration

SQL, Python, operational data workflows, telemetry, analytics, and agentic-system design support the role's need to connect application behavior with trustworthy downstream use.

Cross-functional execution

My record spans manufacturing, engineering, quality, operations, technology, support, leadership, and customer consequence—useful when modernization crosses organizational boundaries.

WHAT I WOULD CONTRIBUTE

- Read legacy functions in the context of operator decisions, production risk, quality, support, and downstream consumers.
- Translate business-critical behavior into explicit application states, interfaces, ownership, diagnostics, and release criteria.
- Connect SQL, historian, and cloud data work through stable naming, timestamps, units, quality, semantics, and stewardship.
- Build reusable modernization patterns without erasing legitimate plant or process differences.
- Keep AI and agentic tools attached to documentation, diagnostics, support, governance, and measurable work.

EXPERIENCE BOUNDARY

> My verified record is strongest in industrial systems, manufacturing operations, SQL/data workflows, modernization mechanisms, and technical translation. It does not include documented years of production Ignition development or AVEVA PI administration. I would enter with that boundary explicit while contributing immediately in systems mapping, data context, operating risk, release discipline, and cross-functional execution.

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Mandate-to-Evidence Alignment

The relevant connection is the operating system around the technology—not keyword substitution.

EVIDENCE MAPPED TO THE WORK

Role mandate	Relevant verified evidence	Immediate contribution
MES and Ignition application modernization	Vape-Jet operating transformation; ERP/Odoo and workflow modernization; Compunetics manufacturing and engineering systems	Map live behavior, dependencies, users, exceptions, support paths, and acceptance conditions before replacement.
SQL-backed manufacturing applications	SQL and Python toolkit; telemetry, dashboards, data workflows, and operational analytics	Connect application logic to schemas, data quality, context, consumers, and operational decisions.
Historian cleanup and stewardship	High-reliability systems, root cause, quality controls, data governance, and consequence-aware change	Inventory usage, ownership, naming, retention, consumer impact, and reversibility before changing the data estate.
Plant-to-cloud integration	Workflow architecture, data products, analytics, automation, and governed agentic systems	Define a usable data contract with semantics, freshness, quality, access, ownership, and consumer acceptance.
Manufacturing and R&D partnership	Cross-functional leadership across operations, engineering, quality, technology, support, and customers	Translate competing needs into explicit decisions, interfaces, priorities, and accountable owners.

OPERATING STANCE

Production first
Application progress is valuable only when operators, quality, maintenance, support, and customer commitments remain protected.

Context travels with data
A tag or event is not useful downstream until its meaning, quality, timestamp behavior, ownership, and asset relationship are stable.

Reusable, not rigid
Standards should simplify support and scaling while preserving process differences that are genuinely necessary.

Learning compounds
Each release should leave behind diagnostics, runbooks, ownership, acceptance evidence, and a better pattern for the next function.

CONTRIBUTION ARC

Days 1-30: understand and stabilize the live system. **Days 31-60:** build and reconcile a bounded target function. **Days 61-90:** release safely, document the pattern, and prioritize the next tranche.

Open the detailed 90-day plan →

ALIGNMENT THESIS

> Bytecubit would gain an engineer-operator who treats Ignition, SQL, historian, and cloud work as one manufacturing system—not as separate technical tickets. That systems view is where my experience is most directly relevant to the mandate.